another. The similarity of the defining traits has led a number of physicists over the years to the strange speculation that black holes might actually be gigantic elementary particles.

In fact, according to Einstein's theory, there is no minimum mass for a black hole. If we crush a chunk of matter of any mass to a small enough size, a straightforward application of general relativity shows that it will become a black hole. (The lighter the mass, the smaller we must crush it.) And so, we can imagine a thought experiment in which we start with ever-lighter blobs of matter, crush them into ever-smaller black holes, and compare the properties of the resulting black holes with the properties of elementary particles. Wheeler's no-hair statement leads us to conclude that for small enough masses the black holes we form in this manner will look very much like elementary particles. Both will look like tiny bundles characterized completely by their mass, force charges, and spin.

But there is a catch. Astrophysical black holes, with masses many times that of the sun, are so large and heavy that quantum mechanics is largely irrelevant and only the equations of general relativity need be used to understand their properties. (We are here discussing the overall structure of the black hole, not the singular central point of collapse within a black hole, whose tiny size most certainly requires a quantum-mechanical description.) As we try to make ever less massive black holes, however, there comes a point when they are so light and small that quantum mechanics does comes into play. This happens if the total mass of the black hole is about the Planck mass or less. (From the point of view of elementary particle physics, the Planck mass is huge —some ten billion billion times the mass of a proton. From the point of view of black holes, though, the Planck mass, being equal to that of an average grain of dust, is quite tiny) And so, physicists who speculated that tiny black holes and elementary particles might be closely related immediately ran up against the incompatibility between general relativity—the theoretical heart of black holes—and quantum mechanics. In the past, the incompatibility stymied all progress in this intriguing direction.

Does String Theory Allow Us to Go Forward?

It does. Through a fairly unexpected and sophisticated realization of black holes, string theory provides the first theoretically sound